The Missing Baseline

Prediction, Intervention, and the Problem of Learning from Changed Outcomes in Higher Education

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Abstract

Predictive analytics in higher education is often evaluated as if prediction and outcome occupy the same observational world. Yet many systems are deployed precisely to change the outcomes they predict: a risk score triggers advising, notification, resource allocation, or another intervention. This paper argues that such deployment creates a missing-baseline problem. Once prediction changes action, post-deployment outcomes no longer directly reveal untreated risk. Successful prevention can weaken the visible statistical relation that justified intervention, while selective decisions make alternative trajectories unobserved. Educational predictive systems should therefore be evaluated as intervention policies with causal histories, not only as estimators scored against their own post-deployment labels.

Keywords: predictive analytics; higher education; performative prediction; early warning; selective labels; causal inference

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Research and assembly record: Appendix A; SOURCE MAP; 000___RETURN_PATH.txt

1 Prediction That Is Supposed to Fail

Educational analytics can be deployed not merely to forecast outcomes but to trigger actions intended to change them. Higher-education predictive systems have been used to identify students judged at risk and target support (Ekowo and Palmer 2016). The Open Academic Analytics Initiative concretely coupled predictive models to intervention strategies intended to improve student outcomes (Jayaprakash et al. 2014). In such settings, the prediction enters the trajectory it will later be used to evaluate.

If a model predicts likely failure and the resulting intervention helps a student succeed, the realized label makes the warning look wrong. Yet the relevant counterfactual is not simply what happened after the warning; it is what would have happened without the warning and the actions it caused. This is a special case of performative prediction: deployment changes the environment that later generates data (Perdomo et al. 2020).

2 The Missing Baseline

The problem becomes acute when deployed models are updated. Liley and colleagues show that prediction scores driving interventions can become victims of their own success: later observations are produced after the earlier model has affected treatment, and naive replacement can introduce bias (Liley et al. 2021). The untreated outcome relation becomes partly counterfactual.

Boeken, Zoeter, and Mooij model this problem as causal domain shift. In alarm-type decision support, outcomes observed under an effective prior model can lead naive retraining to underestimate risk; evaluation must distinguish the deployed policy from a baseline target policy (Boeken et al. 2024).

Together these results support a further inference: successful intervention can produce *institutional forgetting*. Prevention may suppress adverse outcomes so effectively that subsequent training data weaken visible evidence of the risk that originally justified intervention. If an institution learns only from realized post-intervention labels, success can eventually look like evidence that prevention was unnecessary. This is not a claim that every declining risk estimate is an artifact. It is a warning that intervention effect and baseline risk become different quantities once predictions alter action.

3 The Unlived Curriculum

A related problem concerns outcomes that never enter the dataset. Under selective labels, outcomes are observed only after certain decisions, so alternatives not chosen remain unlabeled (Kleinberg et al. 2018). In education, a student steered away from a course produces no grade for that student-course pairing.

Feedback can make this absence self-reinforcing. Prediction-guided allocation can generate more observations where a system already looks, as work on predictive policing demonstrates (Ensign et al. 2018). The educational mechanism is not identical, but it raises the same empirical question: when recommendation changes enrollment, later data partly describe the policy's prior choices.

Kilbertus and colleagues show that selective labels can make the familiar sequence of learning a predictor and applying a deterministic rule inferior to directly learning a decision policy; exploration can be necessary to observe otherwise missing outcomes (Kilbertus et al. 2020). This introduces an ethical cost. Learning about suppressed alternatives may require institutions to vary decisions imposed on real people.

4 Prediction Must Leave a Receipt

If predictions help cause outcomes, the prediction itself can become necessary evidence for later causal reconstruction. Mendler-Dünner, Ding, and Wang show that performative effects can be unidentifiable from ordinary feature-and-outcome records and identify settings in which recording deployed predictions enables estimation (Mendler-D"unner et al. 2022).

For educational systems, this suggests a minimum causal receipt: model version, score or displayed category, recipient, action triggered, timing, and eventual outcome. The point is not maximal data retention. It is to prevent later records from collapsing "low failure because risk was low" with "low failure because the intervention worked." Determining the minimum sufficient receipt is a governance problem because auditability and surveillance burden pull in opposite directions.

5 From Accuracy to Policy Evaluation

An intervention-coupled predictive system has at least three analytically distinct objects:

(1) estimated risk under a stated baseline policy;

(2) the effect of actions triggered by the prediction; and

(3) the outcomes actually observed under deployment.

Collapsing these into a single post-deployment accuracy measure invites category mistakes. A warning can be useful while becoming less literally accurate against treated labels; a replacement model can fit those labels while estimating untreated risk worse (Liley et al. 2021); and causal-domain-shift correction may be required to estimate performance under another policy (Boeken et al. 2024).

Educational systems already contain the institutional ingredients that make this distinction consequential. Degree-planning systems can forecast grades in order to redirect enrollment decisions (Denley 2013); early-alert systems can couple classifications to interventions (Jayaprakash et al. 2014). The proper unit of evaluation is often the coupled prediction-action system rather than the model in isolation.

6 Limits and Unresolved Territory

The argument combines educational deployments with general causal and machine- learning results. That combination does not prove that every educational predictive system exhibits the same feedback. Institution-specific evidence remains necessary: what action a score triggers, whether intervention is delivered, how treatment effects vary, and which baseline policy is meaningful.

Three questions remain especially important. What designs can estimate untreated educational risk without withholding beneficial support? What forms of bounded exploration are ethically legitimate when selective labels hide alternatives? And what is the minimum causal receipt that preserves evaluability without converting every intervention into permanent behavioral surveillance?

7 Conclusion

Predictive analytics becomes conceptually unstable when evaluated as passive forecasting while deployed as active intervention. Warnings and recommendations can change the trajectories that later become labels. The result is a missing baseline: post-deployment data describe a world already altered by earlier predictions.

The practical danger is institutional forgetting. Prevention can erase visible evidence of the risk that made prevention necessary; deterministic decisions can erase outcomes for paths not taken; and missing prediction histories can erase the receipt needed to reconstruct causal effects. Evaluation should distinguish baseline risk, intervention effect, and observed outcome, while preserving enough prediction and action history to keep those quantities meaningfully separable.

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A Assembly and Provenance

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